Module 2: Relations

Discrete Mathematics / Foundations of Computer Science

> **Topics Covered:** Relations, Ordered Pairs, Cartesian Product, Domain, Codomain, Range, Representations (Ordered Pairs, Matrix, Directed Graph), Types of Relations (Empty, Universal, Inverse, Identity), and Properties of Relations (Reflexive, Irreflexive).

1. Introduction to Relations

In mathematics and computer science, we frequently deal with sets of numbers or objects and the connections between them. A foundational concept that describes these connections is the **relation**.

- A relation is simply a connection between two sets of objects. Think of it as a formal rule that pairs elements from one set with elements from another set.
- **Simple Definition:** A relation is a set of ordered pairs that describe a relationship between elements.

Ordered Pair

An ordered pair consists of two elements written in a specific order:

$$(a, b)$$

Here, a is called the **first element**, and b is called the **second element**.

- **Order is crucial:** $(1, 2) \neq (2, 1)$
- **Equality of Ordered Pairs:** Two ordered pairs are equal if and only if their corresponding elements are equal:

$$(a, b) = (c, d) \iff a = c \text{ and } b = d$$

> **EXAMPLE**
>
> If $(x, 4) = (2, y)$, then by element-wise equality:
>
> $x = 2$ and $y = 4$

Cartesian Product

Let A and B be two sets. The Cartesian product of A and B is denoted by $A \times B$ and contains all possible ordered pairs where the first element belongs to A and the second element belongs to B.

EXAMPLE

Let $A = \{1, 2\}$ and $B = \{a, b\}$:

$$A \times B = \{(1, a), (1, b), (2, a), (2, b)\}$$

Cardinality of Cartesian Product:

If set A contains m elements ($|A| = m$) and set B contains n elements ($|B| = n$), then $A \times B$ contains $m \times n$ ordered pairs:

$$|A \times B| = |A| \times |B| = m \cdot n$$

EXAMPLE

If $|A| = 3$ and $|B| = 4$, then:

$|A \times B| = 3 \times 4 = \mathbf{12}$

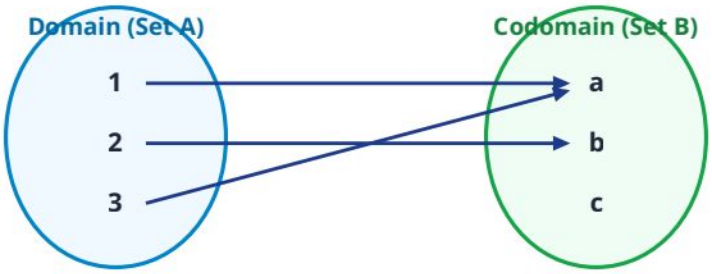
2. Formal Definition of a Relation

Let A and B be two sets. A **relation** R from A to B is defined as any subset of the Cartesian product $A \times B$:

$$R \subseteq A \times B$$

EXAMPLE & MAPPING

Let $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$.
Consider the subset: $R = \{(1, a), (2, b), (3, a)\}$.
Since $R \subseteq A \times B$, R is a valid relation from A to B.



- **Domain:** The set of all first elements appearing in the ordered pairs of the relation.
  Here, $Domain = \{1, 2, 3\}$.
- **Codomain:** The entire second set B that was available to form the relation.
  Here, $Codomain = \{a, b, c\}$.

- **Range:** The set of all second elements actually appearing in the relation.
  Here, $Range = \{a, b\}$.

3. Representations of Relations

1. Ordered Pair Representation

In this method, the pairs of related elements (a, b) are explicitly listed in set notation.

EXAMPLE

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{2, 4, 9, 16, 25\}$.
Let R be the relation of ordered pairs where the second number is the square of the first number:

$$R = \{(a, b) : b = a^2\}$$

Therefore, the ordered pair representation is:

$$R = \{(1, 1), (2, 4), (3, 9), (4, 16), (5, 25)\}$$

2. Matrix Representation

For finite sets, relations can be represented using a boolean zero-one matrix (M_R):

- **Rows** correspond to the elements of the first set (A).
- **Columns** correspond to the elements of the second set (B).
- Put a **1** at row i and column j if $(a_i, b_j) \in R$; otherwise, put **0**.

EXAMPLE

Let $A = \{1, 2, 3\}$ and $B = \{0, 1, 2, 4\}$.
Let $R = \{(1, 1), (1, 2), (1, 4), (2, 2), (2, 4), (3, 4)\}$.

The matrix representation is:

A \ B	0	1	2	4
1	0	1	1	1
2	0	0	1	1
3	0	0	0	1

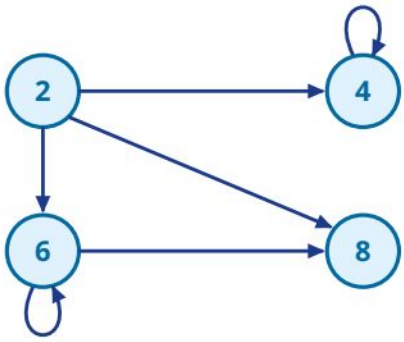
(Rows represent elements from set A · Columns represent elements from set B)

3. Graph Representation (Digraph)

A relation can be represented graphically using a directed graph:

- Each element of the set is represented by a vertex (node).
- A directed arrow is drawn from vertex u to vertex v if $(u, v) \in R$.
- If an element is related to itself, it is represented by a loop.

> **EXAMPLE**
>
> Let $A = \{2, 4, 6\}$ and $B = \{4, 6, 8\}$.
>
> Let $R = \{(2, 4), (2, 6), (2, 8), (4, 4), (6, 6), (6, 8)\}$.



4. Types of Relations

1. **Empty Relation:** A relation R on a set A is called an empty relation if no elements are related to one another:

$$R = \emptyset$$

> **EXAMPLE**
>
> If $A = \{3, 4\}$ and $R = \{(x, y) : x + y > 9\}$.
>
> Since the maximum sum is $4 + 4 = 8 < 9$, no pairs satisfy the condition.
>
> Therefore, $R = \emptyset$.

2. **Universal Relation:** A relation R on a set A is called a universal relation if every element is related to every element of A:

$$R = A \times A$$

> **EXAMPLE**
>
> If $A = \{1, 2, 3\}$, then:
>
> $R = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$.

3. **Inverse Relation:** Let R be a relation from set A to set B. The inverse of R, denoted by R^{-1}, is the relation from B to A defined by:

$$R^{-1} = \{(y, x) : (x, y) \in R\}$$

> **EXAMPLE**
>
> If R = {(1, 2), (3, 2), (5, 1)},
> Then R^{-1} = **{(2, 1), (2, 3), (1, 5)}**.

4. **Identity Relation:** A relation R on a set A is called an identity relation if every element relates *only to itself*:

$$R = \{(x, x) : x \in A\}$$

> **EXAMPLE**
>
> If A = {1, 2}, then R **= {(1, 1), (2, 2)}**.

5. Properties of Relations

1. **Reflexive Relation:** A relation R on set A is called reflexive if every element in A is related to itself:

$$(a, a) \in R \quad \forall\ a \in A$$

> **EXAMPLE**
>
> Let A = {1, 2, 3} and R = {(1, 1), (2, 2), (3, 3), (1, 2)}.
> Since (1, 1), (2, 2), and (3, 3) are all present in R, the relation is **reflexive**.

2. **Irreflexive Relation:** A relation R on set A is called irreflexive if *no* element in A is related to itself:

$$(a, a) \notin R \quad \forall\ a \in A$$

> **EXAMPLE**
>
> Let A = {1, 2, 3} and R = {(1, 2), (2, 3), (3, 1)}.
> There are no self-pairs like (1, 1), (2, 2), or (3, 3) in R.
> Therefore, R is **irreflexive**.

Module 2: Properties of Relations & Functions

Discrete Mathematics Lecture Notes (Continued)

Topics Covered: Properties of Relations (Symmetric, Anti-Symmetric, Transitive) • Introduction to Functions • Function Validity Conditions • Types of Functions (One-to-One / Injective, Onto / Surjective, Bijective).

1. Properties of Relations (Continued)

3. Symmetric Relation

A relation R is called **symmetric** if whenever $(a, b) \in R$, then $(b, a) \in R$.

$$(a, b) \in R \implies (b, a) \in R$$

In words: If a is related to b, then b must also be related to a.

EXAMPLE

Consider the relation:

$$R = \{(1, 2), (2, 1), (2, 3), (3, 2)\}$$

We observe:

- $(1, 2) \in R$ and $(2, 1) \in R$
- $(2, 3) \in R$ and $(3, 2) \in R$

Therefore, the relation is symmetric.

4. Anti-Symmetric Relation

A relation R on a set A is called **anti-symmetric** if:

$$(a, b) \in R \text{ and } (b, a) \in R \implies a = b$$

In words: If a is related to b and b is related to a, then a and b must be the exact same element (i.e., distinct elements cannot be mutually related in both directions).

EXAMPLE

Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (2, 2), (3, 3), (1, 2)\}$.
We have $(1, 2) \in R$, but $(2, 1) \notin R$.
There is no violation of anti-symmetry (the only pairs with their reverse present are the self-pairs where $a = b$).
Hence, the relation is anti-symmetric.

5. Transitive Relation

A relation R is called **transitive** if:

$$(a, b) \in R \text{ and } (b, c) \in R \implies (a, c) \in R$$

In words: If a is related to b, and b is related to c, then a must be related to c.

EXAMPLE

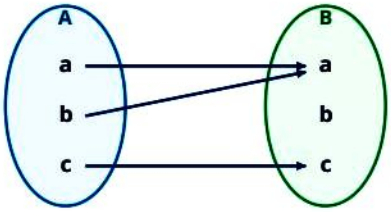
Let R = {(1, 2), (2, 3), (1, 3)}.
We have (1, 2) ∈ R and (2, 3) ∈ R, and the bridged pair (1, 3) ∈ R.
Therefore, the relation is transitive.

2. Functions

Let A and B be two non-empty sets. A relation which assigns **every element of A to a unique element of B** is called a **function** from A to B, written as:

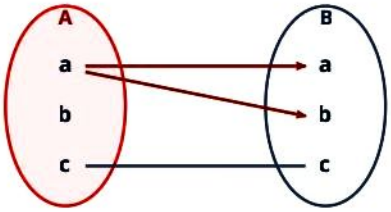
$$f : A \to B$$

Valid Function



$f = \{(a, a), (b, a), (c, c)\}$
Domain(f): $\{a, b, c\}$ • **Range(f):** $\{a, c\}$

Not a Function



Fails function definition:
Element $a \in A$ is assigned to two elements of B (a and b), violating uniqueness.

3. Types of Functions

1. One-to-One (1-1) or Injective Function

A function $f : A \to B$ is said to be **one-to-one (injective)** if different elements in A have distinct images in B:

$$f(x_1) = f(x_2) \implies x_1 = x_2$$

One-to-One (Injective)

Every element of A maps to a unique element of B.

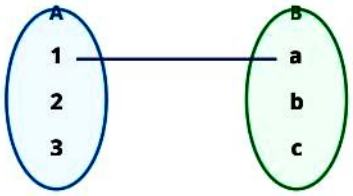
Not One-to-One

Here, a is the image of two elements (1 and 2).

2. Onto or Surjective Function

A function $f : A \to B$ is said to be **onto (surjective)** if each element of B is the image of at least one element of A (i.e., $Range(f) = Codomain(B)$).

Onto (Surjective)



Every element of B has a pre-image in A.

Not Onto

Here, c is not assigned to any element of A.

3. Bijective Function

A function $f : A \rightarrow B$ is said to be **bijective** (or a one-to-one correspondence) if it is **both one-to-one and onto**.

CONCLUSION

This function is **bijective** because:

It is **one-to-one:** Different elements in A have distinct images in B.

It is **onto:** Every element in codomain B has a pre-image in A.